Predictable Scale Dependence Makes Gradient Heavy Tails Removable: Scale-Normalized Sequential Learning

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Abstract

We study online prediction on streams whose loss-gradient *scale* is strongly nonstationary, using a large financial dataset as a testbed. At a fixed predictor the *pooled* gradient-norm tail is genuinely heavy (Hill index $\hat{\alpha} \approx 2.4$)—but it is *conditionally removable*: the *causally scale-normalized* gradient, the quantity a learner actually acts on, has a markedly lighter tail ($\hat{\alpha} \approx 2.9$–3.7, estimator-dependent). This removability is a property of *serial dependence*, not of the marginal: an order-*shuffle* that preserves the marginal exactly destroys it, and the same causal normalization is tail-index *preserving* on iid streams of known index—so the lightening is neither an estimator artifact nor a restatement of the pooled tail. The heavy marginal itself is a volatility-clustering effect: a *bounded* scale drift cannot move a tail index (Breiman, 1965), but a predictable scale with a heavy-tailed conditional law can (the Kesten/GARCH mechanism; Kesten 1973), and a GARCH surrogate with light innovations reproduces both $\hat{\alpha} \approx 2.4$ and its removal. What a learner faces is thus the *predictable* part of the scale, not an intrinsically heavy conditional tail. In this regime plain online gradient descent (OGD) has no usable learning rate—small rates underfit while any rate large enough to learn diverges when the scale spikes—and clipping to a *scale-dependent* threshold is fragile because the clipped step still inherits the drift. We propose *Scale-Normalized OMD* (SN-OMD), which divides the gradient by a tracked robust scale and caps it at a *scale-free* constant M; the per-step move is then bounded by M rather than by a regime-dependent quantity, so the iterates provably cannot blow up—they grow at most as $O(\sqrt{t})$ on $\mathbb{R}^d$ (and stay bounded on a bounded domain) for any learning rate or realization (a stability, not convergence, guarantee). A dynamic-regret analysis—attaining the per-regime rate $T^{1/p}$ under a finite, possibly time-varying p-th moment ($p \in (1, 2]$)—is deferred to the appendix, since on real data the scale drift enters it as a genuine $\sqrt{T}$ factor; we lead instead with the empirically-testable mechanism. Empirically, a whole family of *bounded scale-free* updates—SN-OMD, classical normalized-GD, a constant-threshold clip, and AdaGrad-Norm—stays bounded and accurate across a wide learning-rate range where OGD and scale-dependent clipping diverge, and on this data none dominates: the best per-protocol accuracy belongs to the constant clip and AdaGrad-Norm, not to SN-OMD. What singles out SN-OMD is not accuracy but that its scale is *predictable*, which is exactly what the dynamic-regret guarantee requires; the cap additionally buys non-divergence at large steps.

1. Introduction

Financial markets are a canonical source of heavy-tailed, distribution-free data: returns and microstructure signals exhibit power-law tails, occasional data errors act as adversarial contamination, and assumptions such as sub-Gaussianity are untenable. Sequential prediction on such streams therefore calls for *distribution-free* guarantees that hold under weak moment assumptions only. A rich line of work in robust statistics shows that the empirical mean is a poor estimator under heavy tails, whereas estimators such as Catoni's M-estimator, median-of-means, and the trimmed mean achieve sub-Gaussian deviations assuming only a finite variance (Catoni, 2012; Devroye et al., 2016; Lugosi & Mendelson, 2019); in optimization, gradient clipping is the corresponding algorithmic primitive for heavy-tailed gradient noise (Zhang et al., 2020; Gorbunov et al., 2020; Nguyen et al., 2023).

This paper concerns a facet these lines mostly abstract away: on real streams the gradient *scale* is strongly nonstationary,

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and this—more than an intrinsically heavy tail—is what breaks existing methods. Measuring the gradient of an online linear predictor on the Jane Street market dataset, we find (i) the gradient *scale* drifts roughly $6\times$ across two months with overnight jumps; (ii) this drift alone inflates the *pooled* gradient-norm tail index to $\hat{\alpha} \approx 2.4$—a scale mixture of lighter-tailed increments looks heavy-tailed—whereas the *causally scale-normalized* gradient $\|g_t\|/s_t$, which is what a learner actually acts on, is much lighter, $\hat{\alpha} \approx 2.9$–3.7; and (iii) the *gradient* tail is nonetheless heavier than either its feature or residual factor alone, an *interaction* effect from their co-movement. The operative difficulty is thus nonstationarity of scale, with only a mild residual heavy tail. That a drifting scale manufactures a heavy pooled tail is a classical mechanism for asset *returns* (scale mixtures, ARCH/GARCH, and the near-Gaussianity of volatility-standardized returns; Section 5); our contribution is that it holds for the loss *gradients* of an online learner—as a feature×residual interaction—and that it has an exploitable algorithmic consequence, developed below. This regime—a drifting scale against a moving comparator—is exactly where existing distribution-free online learning results, which are either in expectation, on bounded domains, or against a static comparator (Zhang et al., 2026; Zhang & Cutkosky, 2022), do not reach.

The consequences are sharp. Plain OGD has *no* usable learning rate on such a stream: a small rate underfits, while any rate large enough to learn diverges the first time the gradient scale spikes in a turbulent regime. The textbook fix—clip the gradient at a threshold τ_t tracking the local scale—does not help under a drifting scale, because the clipped step is $\propto \min(\|g_t\|, \tau_t)$ and so still scales *with* the regime: a turbulent window produces large steps and diverges. We show this failure both empirically and as a structural property of any scale-dependent threshold.

Our approach. The fix is to decouple the step from the tail scale. *Scale-Normalized OMD* (SN-OMD) divides the gradient by a predictable robust scale s_t and caps the result at a *scale-free* constant M,

$$\hat{g}_t \;=\; \mathrm{clip}(g_t/s_t,\; M)\,, \qquad w_{t+1} \;=\; \Pi_{\mathcal{W}}\Big(w_t - \tfrac{\eta}{\sqrt{t}}\,\hat{g}_t\Big)\,.$$

The single factor $1/s_t$ replaces clipping's scale-*dependent* step bound Ms_t with a scale-*free* bound M. Because $\|\hat{g}_t\| \le M$ deterministically, the iterates grow at most as $O(\sqrt{t})$ (bounded on a bounded $\mathcal{W}$) for *any* learning rate or realization—an unconditional *stability* guarantee (not, on its own, convergence) that needs no moment assumption.

Contributions.

- **Heavy gradient tails are conditionally removable (Section 4).** On a large market dataset the pooled gradient tail is genuinely heavy ($\hat{\alpha} \approx 2.4$) but a learner with a *predictable* causal scale tracker faces a much lighter one ($\hat{\alpha} \approx 2.9$–3.7). Removability is a property of *serial dependence*: a marginal-preserving order-*shuffle* abolishes it, and the same normalization is tail-index preserving on iid streams of known index (so it is no artifact). The heavy marginal is volatility clustering (rank-based $Q_{100} \approx 7\times10^5$)—a bounded drift cannot produce it (Breiman, 1965), a heavy-tailed predictable scale can (Kesten/GARCH), and a GARCH surrogate with light innovations reproduces both the tail and its removal. The effect recurs, in weaker form, on image-model training gradients. This is the paper's central finding.
- **Algorithm, stability, and a cap frontier (Sections 3 and 4.1).** We propose SN-OMD, prove its iterates cannot blow up ($O(\sqrt{t})$ growth on $\mathbb{R}^d$, bounded on a bounded domain; Theorem 3.1), and show it is a one-parameter family in the cap M interpolating normalized-GD ($M \to 0$) and uncapped scale-adaptive OGD ($M \to \infty$); a two-directional test predicts *when* the finite cap matters—non-divergence at large steps, and an accuracy optimum only when the normalized tail is genuinely heavy.
- **A protocol-careful study, and deferred theory (Section 4).** Under causal standardization SN-OMD stays bounded and accurate (one of several co-leading bounded scale-free methods, not the single best; Table 1) where OGD and scale-dependent clipping diverge; it is invariant to per-row vs. per-timestep updates (no contemporaneous leakage) and its advantage vanishes on a genuinely stationary control. A per-regime dynamic-regret bound (Theorem A.2) is deferred to the appendix, honest about a $\sqrt{T}$ drift factor; a tested implementation is released.

2. Problem Setup

Protocol. Learning proceeds in rounds $t = 1, \dots, T$. At round t the learner plays $w_t \in \mathcal{W} \subseteq \mathbb{R}^d$ (a convex set of diameter D; the unbounded case is discussed in Section 3), observes a convex loss ℓ_t (for us the sample-weighted squared error $\ell_t(w) = \omega_t(\langle w, x_t\rangle - y_t)^2$), and receives a stochastic subgradient g_t with $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = \bar{g}_t \in \partial\ell_t(w_t)$, where $(\mathcal{F}_t)$ is the natural filtration and $\varepsilon_t = g_t - \bar{g}_t$ is the noise.

Drifting heavy-tailed gradients. We assume only a *finite, time-varying* low-order moment of the subgradient itself—the quantity we measure and the algorithm acts on:

Assumption 2.1 (Drifting p-th moment)**.** For each t, $\mathbb{E}[\|g_t\|^{p_t} \mid \mathcal{F}_{t-1}] \le \sigma_t^{p_t}$ for some $p_t \in (1, 2]$, where both the scale σ_t and the tail index p_t may drift with t, and $p_t < 2$ (infinite variance) is permitted. Write $p := \inf_t p_t$. (This bounds $\bar{g}_t$ and the noise $\varepsilon_t = g_t - \bar{g}_t$ up to constants, and is exactly the tail we estimate in Section 4.)

No sub-Gaussianity, boundedness, or stationarity is assumed. *Distribution-free* means the learner knows none of σ_t, p_t, D, or the comparator norm. The regret analysis does additionally require a *predictable* scale tracker (Assumption A.1); we do not treat this as free, and discharge it under a mild drift model (Proposition B.5).

Dynamic regret. Performance is measured against a moving comparator $u_{1:T}$ with path length $P_T = \sum_t \|u_t - u_{t-1}\|$,

$$R_T(u_{1:T}) = \sum_{t=1}^{T} \big[\ell_t(w_t) - \ell_t(u_t)\big] \;\le\; \sum_{t=1}^{T} \langle \bar{g}_t, w_t - u_t \rangle, \tag{1}$$

the notion appropriate for regime-switching streams; static regret is the special case $u_t \equiv u$. We report the sample-weighted zero-mean $R^2 = 1 - \sum_i \omega_i (y_i - \hat{y}_i)^2 / \sum_i \omega_i y_i^2$ on data.

2.1. A budget heuristic for dynamic regret under drift

The achievable rate degrades smoothly with the tail. The core primitive is robust mean estimation: under a finite p-th moment the best estimator of a mean from n samples has error $\Theta(\sigma\, n^{-(1-1/p)})$ (Devroye et al., 2016; Lugosi & Mendelson, 2019), which interpolates the sub-Gaussian rate ($p = 2$) and no concentration ($p \to 1$). Aggregated online this yields, and is matched by, a regret of order $T^{1/p}$ (Zhang et al., 2026):

$$R_T \;\asymp\; \sigma \cdot T^{1/p}, \qquad T^{1/2}\ (p{=}2) \;\longrightarrow\; T\ (p \to 1). \tag{2}$$

Two refinements matter under nonstationarity. Partitioning $[T]$ into regimes r of length L_r, index p_r, and scale σ_r, and applying the bound within each, *static* regret (one comparator) is governed by the worst regime, while *dynamic* regret (a comparator per regime) has the per-regime costs *add*:

$$R_T^{\text{stat}} \gtrsim \max_r \sigma_r L_r^{1/p_r}, \qquad R_T^{\text{dyn}} \gtrsim \sum_r \sigma_r L_r^{1/p_r}. \tag{3}$$

Because $L^{1/p}$ is convex, heavy-tail cost cannot be averaged across regimes: a few short heavy regimes ($p_r < 2$) dominate the sum. Three caveats keep (3) honest. First, it is a per-regime *budget*, not a proved regret lower bound at a fixed path length: it charges each regime its worst-case cost against a *fresh* comparator, so it lower-bounds dynamic regret only against an *unconstrained* comparator sequence—which is trivially $\Omega(T)$ in any case. The interesting object is regret at a given path length P_T, which (3) does not constrain. Second, it separates drift from the static case only qualitatively: prior heavy-tailed results fix a *single* p, a stationary scale, and a *static* comparator, and recover the $T^{1/p}$ rate (2) (Zhang et al., 2026; Zhang & Cutkosky, 2022; Bubeck et al., 2013), whereas here the per-regime costs *add* across a drifting exponent, and the static budget (a max, not a sum) does not see this. Third—and this is why we keep (3) *qualitative*—the sum is dominated by regimes with $p_r < 2$, which we measure only on *pooled* gradients; after scale normalization the tail the learner actually sees has $p \approx 2$ (Section 4), and the super-$\sqrt{T}$ growth of the pooled budget largely evaporates. We therefore read (3) as a *motivating heuristic*—the interaction of tail drift with a moving comparator is what makes the dynamic problem hard—and not as a measured wall; a proper P_T-constrained Ω is left open. What SN-OMD delivers is to attain the per-regime worst-tail rate $T^{1/p}$ rather than diverge.

Algorithm 1 Scale-Normalized OMD (SN-OMD)

Input: cap M, step η, predictable scale tracker $\mathcal{S}$; $w_1 = 0$.
for $t = 1$ **to** T **do**
  play w_t; observe ℓ_t and stochastic subgradient g_t
  $s_t \leftarrow \mathcal{S}.\text{scale}$ (predictable, uses g_t for $t' < t$)
  $\hat{g}_t \leftarrow \text{clip}(g_t/s_t,\ M)$
  $w_{t+1} \leftarrow \Pi_{\mathcal{W}}\big(w_t - \frac{\eta}{\sqrt{t}}\hat{g}_t\big)$
  $\mathcal{S}.\text{update}(\|g_t\|)$
end for

3. Scale-Normalized Online Mirror Descent

Algorithm. SN-OMD maintains a *predictable* robust estimate $s_t > 0$ of the gradient-norm scale (an $\mathcal{F}_{t-1}$-measurable function of past gradient norms) and takes a scale-invariant, constant-capped step (Algorithm 1). Writing $v_t = g_t/s_t$ and $\text{clip}(v, M) = v \cdot \min\{1, M/\|v\|\}$, the update is $w_{t+1} = \Pi_{\mathcal{W}}(w_t - \frac{\eta}{\sqrt{t}}\,\text{clip}(v_t, M))$. The scale s_t can be any predictable robust tracker; we use a winsorized exponential moving average of $\|g_t\|$ by default, and analyze a two-timescale "peak-hold" variant below.

Why normalize, not clip. Clipping to a scale-dependent threshold $\tau_t = c\, s_t$ gives a step $\propto \min\{\|g_t\|, c\, s_t\}$, which scales *with* the drifting s_t: a turbulent regime yields large steps and divergence. Dividing by s_t first gives a step $\propto \min\{\|g_t\|/s_t, M\}$, bounded by the *scale-free* constant M. The two differ by the factor $1/s_t$, and this single factor is what removes the scale-drift transmission.

Theorem 3.1 (Stability, unconditional). *For any inputs, $\|\hat{g}_t\| \le M$ deterministically. Hence on a bounded domain $\|w_t\| \le D$ for all t, and on $\mathbb{R}^d$ (step $\eta/\sqrt{t}$) $\|w_t\| \le \|w_1\| + 2M\eta\sqrt{t}$. Hence the iterates remain uniformly bounded (on $\mathcal{W}$) or grow at most as $O(\sqrt{t})$ (on $\mathbb{R}^d$) for any learning rate, scale drift, or realization—a* stability *guarantee, not by itself convergence, that uses no moment assumption.*

This is the crucial contrast with scale-dependent clipping, whose per-step move $\propto s_t$ is bounded only by the drifting scale.

Dynamic regret (deferred to the appendix). Beyond stability, SN-OMD admits a high-probability *dynamic*-regret analysis (Section A): under a scale-tracking condition—a predictable s_t bracketing the local scale (Assumption A.1)—it attains the per-regime rate $T^{1/p}$ under only a finite p-th moment. We keep it in the appendix, not the headline, for two honest reasons. The bound is stated through the tracker's upward variation $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$, which we *measure* to grow linearly with T on real data (Section 4.3); so the scale drift is a genuine $\sqrt{T}$ factor and the $T^{1/p}$ rate holds per regime, not globally. And the scale-tracking condition, though one-sided and empirically dischargeable (Proposition B.5), remains an assumption. What the body develops instead is the empirically-testable content—why normalizing beats clipping, and where the cap earns its keep.

4. Experiments

Setup and protocol. We use the Jane Street Real-Time Market Data Forecasting dataset (Jane Street, 2024): 47,127,338 rows over 1,699 trading days, 79 features, 39 symbols; target `responder_6`; metric the sample-weighted zero-mean R^2. The stability sweeps (Table 1, Figure 4) stream a continuous 150,000-row slice (the first $\sim$20 trading days). We evaluate *progressive online* prediction on the historical record: each round the learner plays w_t, then observes (x_t, y_t) and updates—the protocol our theory concerns. Features are standardized *causally* (running past-only statistics; a full-sample standardizer inflates R^2 by look-ahead and is not used). We report two round granularities: *per-row* and *per-timestep* (predict the whole $(\text{date}, \text{time})$ cross-section of $\sim$9 symbols, then one aggregated update), the latter carrying no contemporaneous cross-sectional information. **This is not the competition's protocol** (batched timesteps served with day-lagged responders, scored on a held-out future window), so our R^2 is not comparable to competition leaderboard scores in either direction; we report it only for cross-method comparison and the stability contrast. Tail measurements are learner-independent (gradients at the fixed predictor $w^\star$): the pooled index from $\sim$200,000 gradients, the daily-drift index from the full record.

The pooled tail is real, but conditionally removable. At the fixed optimum $w^\star$, the *pooled* gradient-norm tail index is $\hat{\alpha} \approx 2.43$, and it is an *interaction* effect: the residual and feature norms each have $\hat{\alpha} \approx 4$, but their product—the gradient—has a heavier tail because extreme features and residuals co-occur (Figure 1a). This heaviness is genuine, but *conditionally removable*: dividing by a causally-tracked robust scale s_t—the quantity SN-OMD actually acts on—lightens the tail markedly, to $\hat{\alpha} \approx 3.7$ (causal winsorized-EMA s_t), ≈ 3.2 (non-causal median), or $\approx$ 2.9 (trailing median), i.e. toward finite higher moments. What makes it removable is *serial dependence*: the gradient magnitude is strongly clustered—rank (Spearman) autocorrelation ≈ 0.16 at lag 100, rank-based Ljung–Box $Q_{100} \approx 7 \times 10^5$ (ranks are bounded, so the χ^2 is valid; a shuffled control gives $\approx$ 116), of which the $\approx 6\times$ daily drift (Figure 1b) is only the low-frequency component. Three controls (Figure 2; released `research_null_normalization.py`) pin this down: (i) a marginal-preserving order-*shuffle* leaves the pooled index unchanged but destroys the lightening, so the tracker exploits temporal *order*, not the marginal; (ii) the same causal normalization is tail-index *preserving* on iid streams of known index $(2.0, 2.4, 3.0)$, ruling out an estimator artifact; and (iii) a GARCH surrogate with *light* (Gaussian) innovations reproduces both the pooled $\hat{\alpha} \approx 2.4$ and its removal. The heavy *marginal* is *not* manufactured by the bounded $\approx 6\times$ drift—a bounded scale factor cannot move a tail index (Breiman, 1965)—but by a predictable scale whose conditional law is heavy-tailed (the Kesten/GARCH mechanism, Kesten 1973; equivalently the subordination of Clark 1973, Section 5). So the tail a learner faces is set by the *predictable* part of the scale, leaving only a mild residual heavy tail. (Per-day $\hat{\alpha}$ dips below 2 on a minority of days, but single-day Hill estimates are noisy and we do not rely on them.) We note that the per-regime budget of Section 2.1 is dominated by exactly these pooled-$\alpha < 2$ days; since those regimes are themselves a pooled-tail artifact—they largely disappear once the gradient is normalized (the tail the learner sees has $p \approx 2$)—we treat that budget only as qualitative motivation for why a moving comparator under drift is hard, not as a measured quantity.

A second domain, and its limit. On the per-step gradient norm of a small CNN trained on MNIST (2,814 steps), the pooled tail is heavy ($\hat{\alpha} \approx 2.6$) and normalizing lightens it ($\hat{\alpha} \approx 6$; single-window Hill estimates are unstable here, 5–9 across thresholds). But this is *schedule-driven*: within a single constant-learning-rate window (nearly flat scale) the tail is *already* light ($\hat{\alpha} \approx 6$–9), so the pooled heaviness comes from the scale decay and the learning-rate drops, not from within-phase drift. The drift-manufactures-tails effect thus recurs on training gradients, but only in this weaker, transient form; the genuine within-regime version—confirmed by the shuffle and iid-null controls and a feature–residual interaction—we establish only on the market data.

Scale-invariant updates dominate scale-dependent ones. Streaming one learner through the multi-regime window and sweeping the learning rate (Figure 4, Table 1), the methods split cleanly by whether their step responds to the gra-

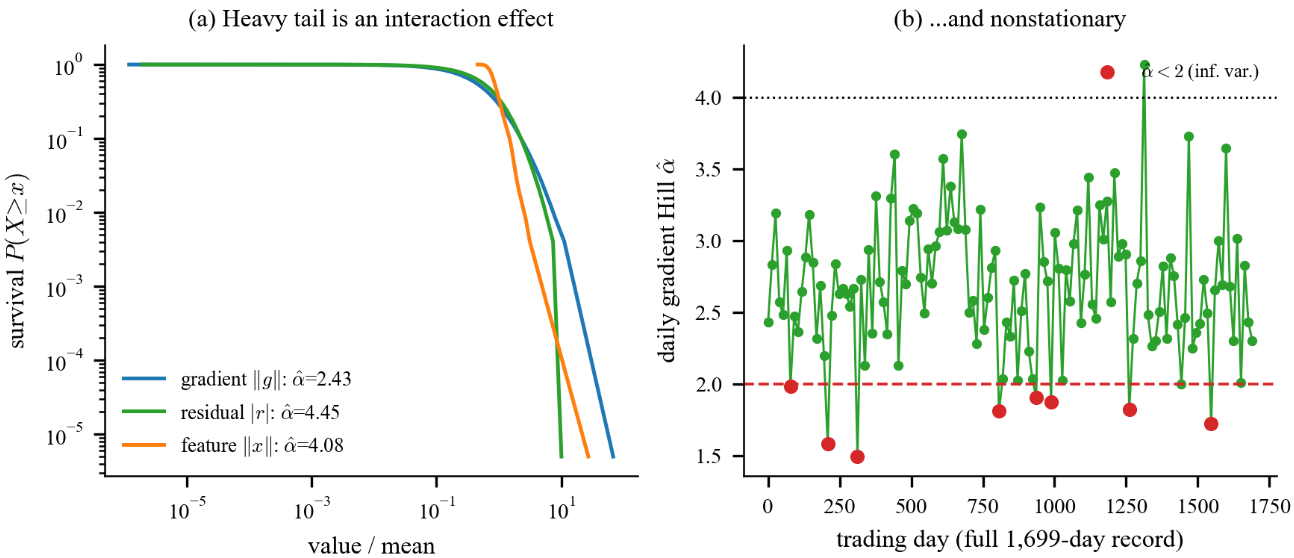

Figure 1. **The pooled tail is real, but conditionally removable.** (a) The *pooled* gradient tail (Hill index $\hat{\alpha} \approx 2.43$) is heavier than either its residual or feature factor ($\hat{\alpha} \approx 4$)—an interaction effect from co-movement. (b) The daily gradient tail index across the full 1,699-day record drifts; per-day estimates occasionally fall below 2 but single-day Hill estimates are noisy. Dividing by a *predictable* causally-tracked scale lightens the tail to $\hat{\alpha} \approx 2.9$–3.7 (estimator-dependent); a marginal-preserving order-shuffle leaves the pooled index unchanged (an order-statistic identity) yet abolishes the lightening, so what the tracker removes is serial dependence, not the marginal (Section 4).

dient *scale*. *Scale-dependent* methods—plain OGD and the robust clippers (AdaptiveClip, RobustOMD), whose threshold $\tau_t = c\, s_t$ rescales with the regime—are fragile: they stay bounded only up to $\mathrm{lr} \approx 10^{-2}$ and reach best $R^2 \approx 0.01$–0.14, because any larger rate diverges when the scale spikes (Figure 3). *Scale-invariant* methods—normalized-GD and SN-OMD (bounded to $\mathrm{lr} = 2$) and uncapped scale-adaptive OGD (to $\mathrm{lr} = 1$)—reach $R^2 \approx 0.08$–0.32 across the two protocols. Per-timestep batching lifts OGD from 0.02 to 0.14 (averaging the $\sim$9 correlated symbols per timestep) but leaves it fragile above $\mathrm{lr} \approx 10^{-2}$; SN-OMD is essentially unchanged ($0.285 \to 0.309$), so its accuracy is not a contemporaneous-information artifact. (Uncapped scale-adaptive OGD instead *halves* under batching, $0.162 \to 0.080$: aggregating ~ 9 gradients enlarges the occasional untruncated step that a finite cap would have contained—a small preview of the cap's role.)

What the tracked scale and the cap buy. Among the scale-invariant methods, SN-OMD and normalized-GD are *comparable* on this data ($0.29/0.31$ per-row/ per-timestep vs. $0.20/0.32$): *normalization*—dividing by the scale—is what earns the gain over the divergent baselines, and normalized-GD is a valid simple instance of it. The same holds for two further scale-free bounded baselines—a constant-threshold fixed-τ clip and AdaGrad-Norm—which are likewise competitive here (each the most accurate entry on one protocol; Table 1), confirming that it is bounded scale-free updating in general, not SN-OMD in particular, that closes the accuracy gap to the divergent methods. SN-OMD's specific contributions lie elsewhere. First, its scale is a *predictable* tracker, which is what makes the regret bound (Theorem A.2) go through; normalized-GD's instantaneous $1/\|g_t\|$ is not predictable and discards gradient magnitude every round. Second, the cap buys *non-divergence at large steps*: uncapped scale-adaptive OGD diverges at $\mathrm{lr} \geq 2$ (R^2 collapses to -63), whereas SN-OMD ($M{=}5$) stays bounded there and in fact peaks at $\mathrm{lr} = 2$. The two are close on Jane only because the *normalized* tail is mildly heavy ($\hat{\alpha} \approx 3.7$); Section 4.1 shows that when it is genuinely heavy the cap's accuracy edge becomes decisive.

Negative control. If the advantage is really about scale drift, it should vanish when the scale is constant. Imposing genuine stationarity in the synthetic harness ($\sigma_t \equiv 1$, $p{=}2$, static reachable comparator—the one stream where stationarity can be enforced), tuned SN-OMD is if anything *worse* than tuned OGD (steady excess loss 6.6 vs. $5.6{\times}10^{-4}$ over 400 seeds, an 18% gap the wrong way): the advantage disappears as the thesis predicts. (A 100k-row Jane slice is *not* a valid stationarity control—it still has $\sim 30\%$ within-day drift—so SN-OMD's lead there is expected; what it does isolate is that the clippers' tail-rejection, unlike scale-invariance, does not even beat OGD, 0.013–0.015 vs. 0.02.)

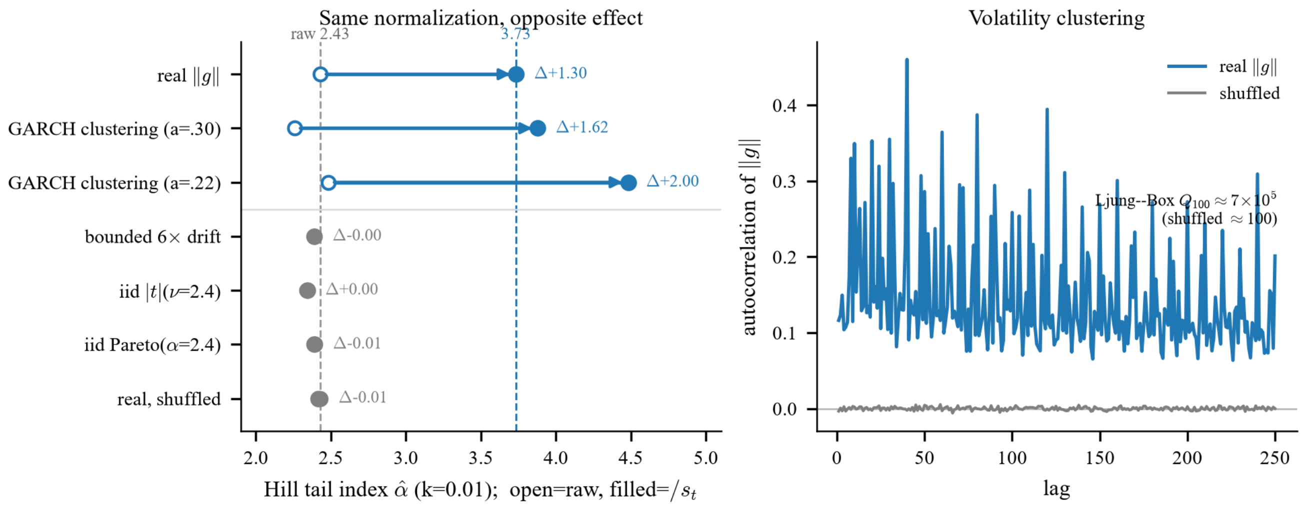

Figure 2. **The lightening is removability, not manufacture.** (a) Hill index before (open) and after (filled) causal scale-normalization, k=0.01. The *same* operation lightens the real gradient tail (2.43 → 3.73) and a GARCH surrogate with light innovations ($\approx$ 2.3 → 3.9–4.5), but leaves an order-*shuffle* of the real series, iid heavy streams of known index, and a bounded 6× drift essentially unchanged ($\Delta \approx 0$)—so what is removed is serial dependence, not the marginal, and it is not an estimator artifact. (b) Rank autocorrelation of $\|g_t\|$ is slowly-decaying (the volatility clustering the tracker exploits) and vanishes under shuffling.

4.1. The cap M interpolates normalized-GD and scale-adaptive OGD

SN-OMD is a one-parameter family in the cap M: when $\|g_t\|/s_t > M$, $\mathrm{clip}(g_t/s_t, M) = M\, g_t/\|g_t\|$ and s_t cancels, so $M \to 0$ is *normalized-GD* and $M \to \infty$ is uncapped *scale-adaptive OGD*. Every *finite*-M member satisfies Theorem 3.1 ($\|\hat{g}_t\| \le M$)—the normalized-GD endpoint included, its step being $M\, g_t/\|g_t\|$; the uncapped $M \to \infty$ endpoint is the lone exception ($\|\hat{g}_t\| = \|g_t\|/s_t$ is unbounded) and is exactly the member that diverges (Table 1). So against the *other* stable member, normalized-GD, non-divergence is not what singles out SN-OMD; the point is that a finite M *preserves gradient magnitude within the bulk* (the step shrinks near a good predictor) and truncates only the tail, whereas normalized-GD discards magnitude every round.

Sweeping M (each cap at its own tuned step, error bars over seeds) turns the cap into a *mechanism test*. Normalization already removes most of the tail ($\hat{\alpha}$: 2.4 → 2.9–3.7), so the cap should earn its keep only when the *normalized* gradient is still heavy—a prediction with opposite consequences in two settings. On a synthetic stream with an *imposed* heavy normalized tail ($p = 1.5$, static reachable comparator; Figure 5) an interior optimum appears: the best cap is $\approx$ 240× better than the uncapped end (heavy-tail instability) and $\approx$ 10× better than the normalized-GD end (magnitude-blind oscillation), both outside the seed-noise band. On Jane, where the normalized tail is only mildly heavy, there is *no* meaningful interior accuracy optimum: normalized-GD, SN-OMD, and uncapped scale-adaptive OGD are comparable (Table 1). Same mechanism, both predictions borne out. This gives the cap a precise scope: it buys *accuracy* when the tail survives normalization, and *stability unconditionally*—uncapped scale-adaptive OGD diverges at lr $\ge$ 2 while SN-OMD does not (Table 1).

4.2. Synthetic streams with known (σ_t, p_t)

On synthetic streams with a controlled tail (Student-t index p; Figure 7 in the appendix), plain OGD's fitted cumulative-regret exponent rises $0.37 \to 0.70 \to 1.55$ as p falls $2 \to 1.3$—super-linear (divergent) once the tail is heavy—reproducing the real-data divergence with a *known* tail index, so the tail is the mechanism, not a learning-rate artifact. SN-OMD stays sublinear, with exponents *below* $1/p$ (the stochastic stationary instance is easier than the worst case, so this is consistent with, not a tightening of, Theorem A.2); against a drifting comparator its dynamic regret grows with path length at exponent ≈ 0.43, consistent with the $\sqrt{DP_T}$ term.

4.3. Discharging the scale-tracking assumption

The regret bound (Theorem A.2) rests on Assumption A.1, that a *predictable* scale s_t brackets the local noise scale. We close this loop empirically on the real gradient-scale process (200,000 gradient norms at $w^\star$, ground-truth scale a centered rolling median, all trackers predictable), where the true upward variation is V_σ^+. The relevant requirement is one-sided (Section 3): hold the lower bracket $s_t \ge \frac{1}{2}\sigma_t$ everywhere, and keep the upward variation W_s (which enters Theorem A.2 as $\sqrt{W_s}$) as close to the true drift V_σ^+ as possible. A single-timescale trailing median cannot do both: a fast window tracks the scale but *churns* on heavy-tailed

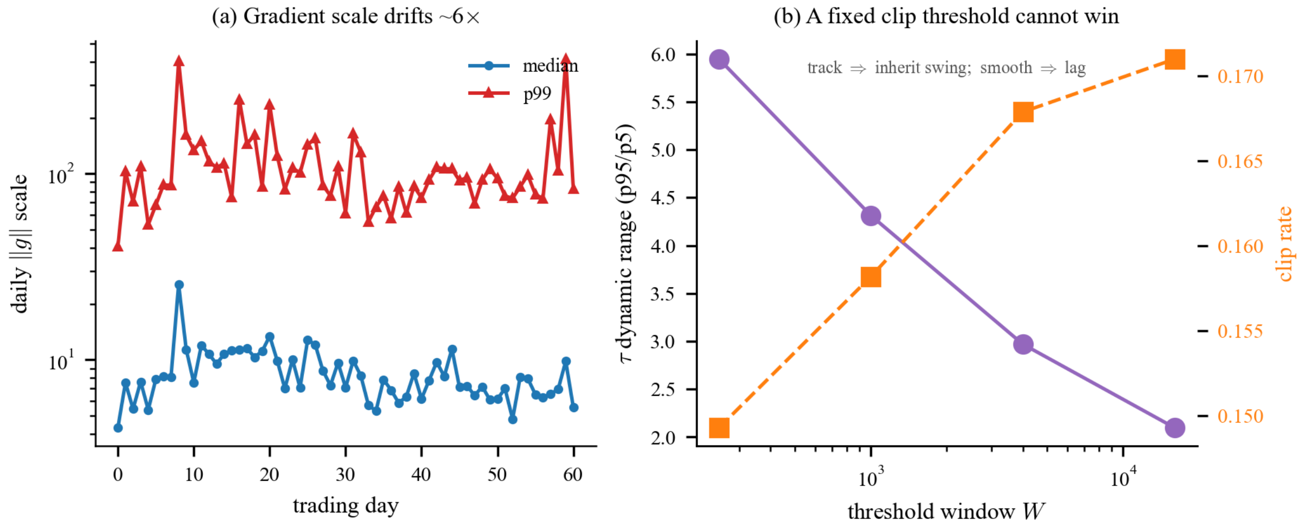

Figure 3. **Why scale-dependent clipping fails.** (a) The gradient scale drifts $\sim 6\times$ across days. (b) A fixed-window adaptive clip threshold cannot win: a short window tracks the scale (so the step inherits the swing), a long window is smooth but lags regime onsets.

spikes ($W_s = 62\, V_\sigma^+$), while a slow window is smooth but *lags* regime onsets and dips below the lower bracket. A two-timescale "peak-hold" envelope—react up on a short robust median, release down only geometrically—resolves the tension: it holds the stability-critical lower bracket at 100% of rounds while attaining the *smallest* upward variation, $W_s \approx 6.6\, V_\sigma^+$ (Figure 6, Table 2). Its only cost is a benign 24% over-estimation of the scale (mild underfit, never divergence). So the lower-bracket half of Assumption A.1 is discharged, and the envelope tracks the true drift to within a small factor ($W_s \approx 6.6\, V_\sigma^+$).

The drift is a genuine rate; the tracker controls only its constant. W_s enters Theorem A.2 as $\sqrt{W_s}$, so it matters how W_s grows with T. It grows *linearly*: W_s sums positive increments over the tracker's update events, and for $100 \le B \le 3000$ plotting it against the update count gives a straight line with a slope independent of the block length ($W_s \approx 1.7\,(T/B)$, linear-fit $R^2 \approx 0.98$–0.99; the slope persists (≈ 1.6) at $B{=}10^4$ but the fit is noisier there, $R^2 \approx 0.87$ on only 20 blocks; $B{=}1$ is a $\sim 5.7\times$ per-round-jitter outlier). So $\sqrt{W_s} \sim \sqrt{T}$ is a genuine rate, and Theorem A.2's $T^{1/p}$ is a *per-regime* statement (globally the drift adds a $\sqrt{T}$ factor). (A naive $W_s \sim T^\beta$ fit appears to give β falling to 0.17 at large B, but that is a finite-horizon artifact: restricting the fit to horizons with $\gtrsim 50$ blocks gives $\beta \approx 1.0$–1.25 with no monotone trend, and $B{=}10^4$ has no such horizon—only ~ 20 blocks fit in our 200k-round record.) What the update timescale *does* control is the constant $1/B$: a regime-timescale tracker shrinks W_s from $\sim 2\times 10^6$ (round-level) to ~ 30 ($B \approx 10^4$, $\approx$ one day; Table 3)—a $\sim 250\times$ smaller $\sqrt{W_s}$ at our horizon, keeping the bound useful in practice. This is Remark A.4's $\rho \asymp 1/L$ as constant control, not exponent control; escaping the $\sqrt{T}$ would require a switching bound in the regime count N (Section B.7), which we leave open. The lower bracket dips slightly (to 0.92–0.93, Table 3) at the block lengths that shrink W_s—trading a little of the stability-critical lower bracket for the smaller constant. This dip is tolerable within the tuned operating range: running the $B{=}10^4$ tracker through the learning-rate sweep, SN-OMD stays bounded through $\mathrm{lr} \le 2$ (peak rolling loss 3.9 at lr=2, versus 8.1 for the winsorized-EMA default at the same rate). We stress-tested this configuration rather than banking its flattering headline number. Two things survive and one does not. *(i) Against the normalized-GD baseline the block tracker is genuinely better per-row*: $R^2 = 0.38$ vs. 0.197 at each method's tuned rate, a paired 10-fold gap of $+0.176$ that is tight and significant ($t \approx 10.6$, all 10 folds), so the "ties normalized-GD" concession (Table 1) is *over-generous for this configuration* on the per-row protocol. *(ii) But the advantage is protocol-specific*: under the honest batched (per-timestep) protocol it evaporates—block 0.351 vs. normalized-GD 0.316, a paired gap of -0.009 ($t \approx -0.1$), indistinguishable—so the headline tie still stands where it is measured most conservatively. *(iii) The edge over the EMA* default *is not significant* either way ($+0.16$ per-row but $t \approx 1.3$, 8/10 folds; not distinguishable at this sample size). Past the tuned range both trackers *degrade in loss* but neither diverges in the sense of Theorem 3.1: at lr=10 the peak rolling loss grows to ≈ 30 (block) vs. ≈ 230 (EMA) while the $O(\sqrt{t})$ iterate bound still holds—it is the loss, not the iterate, that grows, unlike scale-dependent OGD's genuine blow-up. So "does not destabilize" holds within tuning, and within it the block tracker degrades far more gracefully.

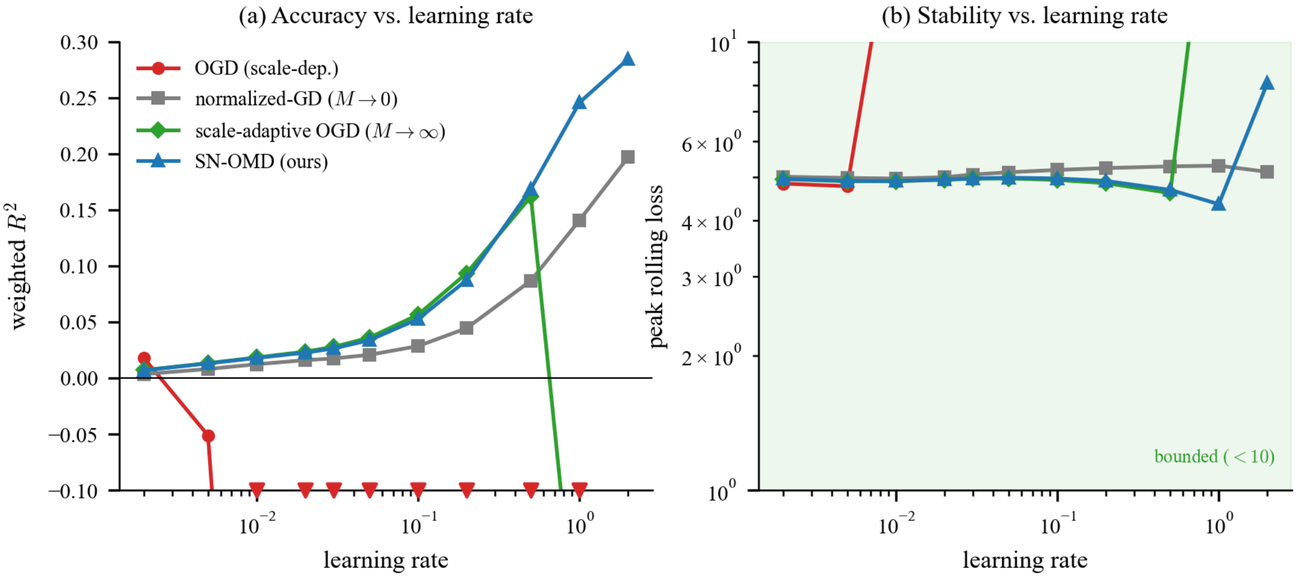

Figure 4. **Main result** (continuous multi-regime stream, *causal* standardization, per-row). The cap family: OGD (scale-dependent) vs. the $M \to 0$ endpoint (normalized-GD), the $M \to \infty$ endpoint (uncapped scale-adaptive OGD), and SN-OMD ($M{=}5$). (a) Weighted R^2 vs. learning rate: OGD cliffs above $\mathrm{lr} = 10^{-2}$ (off-scale), the scale-invariant methods climb to $R^2 \approx 0.2$– 0.3; SN-OMD and normalized-GD are comparable here (the per-row point estimate is slightly higher for SN-OMD but within fold noise, $t \approx 0.1$; Table 1), and uncapped scale-adaptive OGD diverges at large rates. (b) Peak rolling loss: OGD explodes early and uncapped scale-adaptive OGD at large rates (10^2–10^{30}); normalized-GD's peak loss stays ≈ 5 across the range and SN-OMD's stays single-digit, rising to ≈ 8 at its top stable rate lr=2.

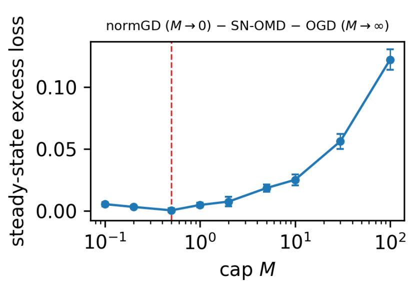

Figure 5. **The cap has an interior optimum** (synthetic, $p = 1.5$). Steady-state excess loss vs. M, each cap at its own tuned step: the finite-M optimum beats both the normalized-GD end ($M \to 0$) and the uncapped scale-adaptive-OGD end ($M \to \infty$).

5. Related Work

Scale mixtures and volatility models. That a drifting scale makes conditionally light-tailed data look unconditionally heavy-tailed is classical *for asset returns*: Clark's mixture-of-distributions / subordinated-process model (Clark, 1973), the ARCH/GARCH family that lets the conditional variance vary in time (Engle, 1982; Bollerslev, 1986), and the finding that returns *standardized by realized volatility* are approximately Gaussian (Andersen et al., 2001). Our pooled-vs-normalized separation is the same phenomenon, and we claim no novelty for the mechanism itself. What is new is (i) that it holds for the *loss gradients* of an online learner—not the returns—so it governs *optimization*, not just modeling; (ii) that the gradient's heavy tail is an *interaction* effect (feature × residual co-movement), heavier than either factor alone (Figure 1); and (iii) the algorithm-design consequence for online learning—the scale-normalized update and its cap-interpolation frontier (Section 4.1)—which the volatility literature does not address.

Robust estimation and clipping. That the empirical mean fails under heavy tails, and robust estimators (Catoni, median-of-means, trimmed mean) achieve sub-Gaussian deviations under a finite variance, is classical (Catoni, 2012; Devroye et al., 2016; Lugosi & Mendelson, 2019); Bubeck et al. (2013) import this into sequential decision-making. In optimization, gradient clipping yields high-probability convergence under heavy-tailed noise (Zhang et al., 2020; Gorbunov et al., 2020; Nguyen et al., 2023). Our normalization is a streaming, scale-decoupled variant whose value is specifically stability under a *drifting* scale.

Heavy-tailed online learning. Zhang et al. (2026) show plain OGD is optimal *in expectation* on a bounded domain, and Zhang & Cutkosky (2022) give parameter-free *high-probability* static regret $\tilde{O}(\|u\|T^{1/p})$ over unbounded domains, whose central obstacle—exponentially large iterates—our cap sidesteps. Both are static; the dynamic, nonstationary regime we target sits outside them,

Table 1. Best weighted R^2 (each method at its own tuned rate) on the continuous Jane Street stream, *causal* standardization, for both round granularities, with the largest stable learning rate; each R^2 carries $\pm$ one circular block-bootstrap standard error (block $\approx$ one trading day, 4000 resamples). Scale-*dependent* methods (top) are fragile (stable only to $\mathrm{lr} \approx 10^{-2}$); scale-*invariant* methods (bottom) stay bounded much higher (SN-OMD and normalized-GD to $\mathrm{lr} = 2$, uncapped only to $\mathrm{lr} = 1$) and are far more accurate. SN-OMD and normalized-GD are *comparable* here; the cap's role is non-divergence (uncapped diverges at $\mathrm{lr} \geq 2$). Two further scale-free bounded baselines—a constant-threshold *fixed-τ clip* (τ=20) and *AdaGrad-Norm*—both satisfy Theorem 3.1 and are accuracy-competitive, each the most accurate entry on one protocol (fixed-τ per-step 0.382, AdaGrad-Norm per-row 0.329); this supports the thesis that *bounded scale-free* updating, not SN-OMD specifically, is what earns the accuracy over the divergent baselines, SN-OMD's distinct contributions being its *predictable* tracker (Theorem A.2) and non-divergence at large steps. The constant threshold cannot track the $\sim 6\times$ scale drift, so its peak loss still grows with the rate (usefully stable only to $\mathrm{lr} \approx 0.2$), whereas AdaGrad-Norm's monotone accumulator keeps its step $\leq \eta$. R^2 is not comparable to competition scores (different protocol, see text). The reported SN-OMD row uses the winsorized-EMA tracker; an alternative block-median tracker changes the per-row number but not per-step, and not beyond fold noise against this row (Section 4.3).

Method	R^2 per-row	R^2 per-step	Stable lr
Scale-dependent (fragile):			
OGD	$0.018_{\pm.006}$	$0.139_{\pm.012}$	$\leq 10^{-2}$
AdaptiveClip / RobustOMD	~ 0.01	—	$\leq 10^{-2}$
Scale-invariant (bounded; stable lr per row):			
Normalized-GD ($M \to 0$)	$0.197_{\pm.024}$	$0.316_{\pm.021}$	≤ 2
Scale-adaptive OGD ($M \to \infty$)	$0.162_{\pm.022}$	$0.080_{\pm.013}$	≤ 1
Fixed-τ clip (const M)	$0.246_{\pm.046}$	$\mathbf{0.382}_{\pm.047}$	≤ 0.2
AdaGrad-Norm	$\mathbf{0.329}_{\pm.030}$	$0.219_{\pm.014}$	≤ 10
SN-OMD (M=5, ours)	$0.285_{\pm.079}$	$0.309_{\pm.056}$	≤ 2

and our budget heuristic (Section 2.1) suggests why it is harder—though a proper P_T-constrained lower bound remains open.

Dynamic, scale-free, and adaptive OL. Dynamic regret against a moving comparator goes back to Zinkevich (2003); scale-free methods adapt to unknown gradient scale (Orabona & Pál, 2018); strongly-adaptive and parameter-free wrappers adapt to unknown interval/regime structure (Daniely et al., 2015; Cutkosky, 2020). We use the last as a black box to remove knowledge of the regime length, and show SN-OMD supplies the required interval-regret interface.

Adaptive optimizers. Coordinate-wise adaptive methods (AdaGrad, RMSProp, Adam) also rescale the gradient by a running second-moment estimate and superficially resemble our normalization, so a natural question is whether they already provide robustness to a drifting scale. The answer splits by whether the preconditioner is *monotone*. For the *EMA-denominator* members (RMSProp, Adam) it is no: their preconditioner divides by $\sqrt{\mathrm{EMA}(g_t^2)}$, which *itself* spikes with a heavy-tailed gradient (the same round enters numerator and denominator), and with no scale-*free* cap a single extreme gradient still yields an unbounded step—so adaptive rescaling alone does not prevent the divergence documented in Section 4. *AdaGrad-Norm*, however, divides by the *monotone* accumulator $\sqrt{\sum_s \|g_s\|^2}$, which bounds its step by η and so does not diverge; run head-to-head on the Jane stream (Table 1) it stays bounded and is accuracy-competitive with SN-OMD (indeed best per-row), placing it with the scale-free bounded family rather than the divergent one. What it lacks relative to SN-OMD is a *predictable* scale (its denominator depends on the current round) and the accompanying dynamic-regret guarantee (Theorem A.2). Our unconditional stability (Theorem 3.1) still comes precisely from capping the *normalized* gradient at a constant M.

6. Conclusion

On real streams the difficulty is dominated by a strongly *non-stationary* gradient scale—which alone makes the pooled gradient look heavy-tailed, though the scale-normalized gradient is much lighter—and a budget heuristic (Section 2.1) motivates why a moving comparator under such drift is hard (we do not lean on it: its super-$\sqrt{T}$ growth is driven by pooled-tail regimes that vanish after normalization). The right response is to decouple the update from the tail scale: SN-OMD normalizes the gradient by a tracked scale and caps it at a scale-free constant, so its iterates provably cannot diverge—at most $O(\sqrt{t})$ for any learning rate—and it attains the worst-tail rate $T^{1/p}$ *per regime*. The scale drift enters as $\sqrt{W_s}$, and W_s grows linearly with T on a persistently fluctuating scale—so the drift is a genuine rate, not a free constant, and the regret is $T^{1/p}$ only per regime; a regime-timescale tracker shrinks the constant but not the exponent (Section 4.3). Empirically SN-OMD's iterates stay bounded across the whole tested learning-rate range where OGD and scale-dependent clipping diverge—accuracy degrading gracefully past the tuned range rather than blowing up (Section 4.3)—and it matches the classical normalized-GD in accuracy while adding a regret guarantee and non-divergence at large steps.

Limitations. SN-OMD is not fully tuning-free: its best learning rate is setting-dependent, and its cap M, decay, and winsorization are untuned defaults—though its stable learning-rate range is far wider than the scale-dependent baselines' and it degrades gracefully. The proven $W_s = O(V_\sigma^+)$ guarantee is for the two-timescale envelope tracker; the deployed default is a winsorized EMA and the most accurate per-row on our stream is a block median, neither of which Proposition B.5 covers (Section 4.3). Closing that gap with a single tracker that is both provably bounded-variation and empirically best is open, as is making the envelope's release rate ρ adaptive to an unknown regime length. So the dynamic-regret bound is proven only for the envelope variant (Assumptions A.1 and 2.1); the deployed EMA's high-probability regret under drift is supported empirically. Empirically, three gaps remain: the EMA-denominator adaptive optimizers (RMSProp, Adam) are argued to diverge but not yet run head-to-head, though the monotone AdaGrad-Norm and a constant-threshold clip now are (Section 5 and Table 1); evaluation centers on accuracy and stability rather than mechanism-level diagnostics; and validation beyond our two domains is warranted.

Software and Data

The released code provides the `ScaleNormalizedOGD` learner as a library class, the scale-tracker implementations used in the experiments (winsorized EMA, two-timescale envelope, and block median) as experiment scripts, the configs, and the scripts that reproduce every figure and number. The primary data is the public Jane Street Real-Time Market Data Forecasting competition; the second-domain gradient norms are logged from a small CNN trained on MNIST (Section 4).

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Dynamic-regret analysis

Beyond the unconditional stability of Theorem 3.1, SN-OMD admits a dynamic-regret bound under a condition on the scale tracker. We state it here (the proof is in Section B) and are explicit about its two soft spots—an assumption on the tracker, and a drift factor we measure to be a genuine rate on real data—which is why the body leads with the empirical results instead.

Assumption A.1 (Scale tracking)**.** The predictable scale brackets the local noise scale: $c_1\sigma_t \le s_t \le c_2\sigma_t$ for constants $0 < c_1 \le c_2$.

Only the lower bracket is essential—it bounds $\|v_t\|$ and hence the rate, and under-estimating the scale is the sole divergence risk—while over-estimation merely shrinks the step η/s_t into mild underfitting, never divergence. Assumption A.1 is thus *one-sided*, which is what makes it dischargeable rather than an oracle (Proposition B.5, and Section 4.3 empirically). Define the tracker's upward variation $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$ and the scale-weighted path $P_T^s = \sum_t s_t\|u_t - u_{t-1}\|$.

Theorem A.2 (Dynamic regret)**.** *Under Assumptions A.1 and 2.1, SN-OMD with $M = \Theta(T^{1/p})$ (or a doubling schedule, requiring no knowledge of p) satisfies, for any fixed comparator sequence $u_{1:T}$, with probability $\ge 1-\delta$,*

$$\begin{aligned} R_T(u_{1:T}) &= O\Big(\big(D\sqrt{W_s} + \sqrt{D\,P_T^s}\,\big)\sqrt{S_1}\;T^{(2-p)/(2p)}\Big) \\ &= \tilde{O}\Big(\big(\sup_t \sigma_t\big)\,\big(D + \sqrt{DP_T}\big)\,T^{1/p}\log(1/\delta)\Big), \end{aligned} \tag{4}$$

where $S_1 = \sum_t \sigma_t$. The second line additionally assumes bounded *scale variation ($W_s = O(1)$, with $S_1 \le T\sup_t \sigma_t$); **this assumption fails on our data**, where $W_s \sim T$ (Section 4.3), so the $T^{1/p}$ form is per-regime—see the remark below.*

Remark A.3 (The drift is a genuine rate). The tail index p sets the exponent (through $M^{2-p} = T^{(2-p)/p}$, giving $T^{1/p}$), while the scale drift enters through $\sqrt{W_s}$. On a persistently fluctuating scale W_s accumulates *linearly* in T ($W_s \sim T$, verified in Section 4.3), so $\sqrt{W_s} \sim \sqrt{T}$ is a genuine rate: (4) is $T^{1/p}$ *per regime*, or $T^{1/p+1/2}$ over a horizon with a growing number of regimes—which at $p = 2$ is the *vacuous* T, so the theorem's content is strictly per-regime. The tracker's update timescale controls the *constant* in W_s (Remark A.4), not the exponent. Escaping the $\sqrt{T}$ would need a switching bound in the regime count, left open.

Remark A.4 (Tracker design). A peak-hold envelope $s_t = \max\{c \cdot \mathrm{median}(\text{recent }\|g_t\|), (1-\rho)s_{t-1}\}$ has $W_s \approx \sup_t s_t + \rho\sum_t s_t$; choosing $\rho \asymp 1/L$ (the regime length) minimizes the W_s constant, whereas too large a ρ churns. An unknown regime length is handled by a strongly-adaptive wrapper (Daniely et al., 2015; Cutkosky, 2020) using SN-OMD's interval-regret property (Theorem 3.1 gives the required bounded feedback).

Remark A.5 (Scope: comparator and step schedule). Two limitations are worth stating plainly. *(i) Comparator.* The high-probability guarantee is per *fixed* (predictable) comparator sequence $u_{1:T}$: the martingale term needs u_t to be $\mathcal{F}_{t-1}$-measurable (Lemma B.3), so the Freedman event is comparator-specific and Theorem A.2 does not hold uniformly over all comparators without a union/chaining bound. In particular the *per-round minimizer* $u_t = \arg\min_u \ell_t(u)$ is $\mathcal{F}_t$-measurable, not predictable, and is *not* covered; our dynamic comparator must be fixed in advance with a given path length. *(ii) Step schedule.* Theorem A.2 is proven for the tuned constant step $\eta_t \equiv \eta$ (known horizon T, or a doubling schedule), whereas Algorithm 1 and Theorem 3.1 deploy the anytime $\eta_t = \eta/\sqrt{t}$ for horizon-free stability. These are the same update under two schedules; carrying the fixed-η regret to the $1/\sqrt{t}$ schedule is the standard anytime/doubling reduction, which we do not reproduce here.

B. Proofs

Throughout, $v_t = g_t/s_t$, $\hat{g}_t = \mathrm{clip}(v_t, M)$ with $M \ge 1$, and $p = \inf_t p_t$. We use the filtration $(\mathcal{F}_t)$ with $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = \bar{g}_t$ and s_t predictable ($\mathcal{F}_{t-1}$-measurable).

B.1. Stability (Theorem 3.1)

Since $\hat{g}_t = v_t \min\{1, M/\|v_t\|\}$, $\|\hat{g}_t\| \le M$ for every realization. On a bounded domain $\Pi_{\mathcal{W}}$ keeps $\|w_t\| \le D$. On $\mathbb{R}^d$, $\|w_{t+1} - w_t\| \le (\eta/\sqrt{t})M$, so $\|w_t\| \le \|w_1\| + M\eta\sum_{k<t} k^{-1/2} \le \|w_1\| + 2M\eta\sqrt{t}$. No moment assumption is used. □

B.2. Clipped moment and bias

Lemma B.1. *For $v \in \mathbb{R}^d$, $M > 0$, $p \in (1, 2]$: $\|\mathrm{clip}(v, M)\|^2 \le M^{2-p}\|v\|^p$ and $\|v - \mathrm{clip}(v, M)\| \le \|v\|^p/M^{p-1}$.*

Proof. If $\|v\| \le M$: $\mathrm{clip}(v, M) = v$ and $\|v\|^2 = \|v\|^p\|v\|^{2-p} \le \|v\|^p M^{2-p}$, while $v - \mathrm{clip}(v, M) = 0$. If $\|v\| > M$: $\|\mathrm{clip}(v, M)\|^2 = M^2 = M^p M^{2-p} \le \|v\|^p M^{2-p}$, and $\|v - \mathrm{clip}(v, M)\| = \|v\| - M \le \|v\| = \|v\|^p\|v\|^{1-p} < \|v\|^p M^{1-p}$. □

Lemma B.2 (Moment and bias under Assumption A.1)**.** *There is a constant κ with $\mathbb{E}[\|\hat{g}_t\|^2 \mid \mathcal{F}_{t-1}] \le M^{2-p}\kappa$, and the bias $b_t = \mathbb{E}[v_t - \hat{g}_t \mid \mathcal{F}_{t-1}]$ obeys $\|b_t\| \le \kappa/M^{p-1}$.*

Proof. Take conditional expectations in Lemma B.1 with $v = v_t$ and use $\mathbb{E}[\|g_t\|^{p_t} \mid \mathcal{F}_{t-1}] \le \kappa_0\sigma_t^{p_t}$ and $s_t \ge c_1\sigma_t$: $\mathbb{E}\|\hat{g}_t\|^2 \le M^{2-p}\mathbb{E}\|v_t\|^p \le M^{2-p}\kappa_0 c_1^{-p}$, and $\|b_t\| \le \mathbb{E}\|v_t\|^{p_t}/M^{p_t-1} \le \kappa_0 c_1^{-p_t}/M^{p-1}$ (using $M \ge 1$, $p_t \ge p$). Set $\kappa = \kappa_0 c_1^{-2}$. □

Because $M \ge 1$ and $p_t \ge p$, all uses of $M^{2-p_t}, M^{-(p_t-1)}$ above are bounded by the worst-case $M^{2-p}, M^{-(p-1)}$; the single-p bounds are thus valid for drifting p_t.

B.3. High probability

Lemma B.3 (Freedman step)**.** *Let (X_t) be a martingale difference sequence with $|X_t| \le b$ and $\sum_{t\le T}\mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}] \le V$ almost surely. Then w.p. $\ge 1-\delta$, $\sum_{t\le T} X_t \le \sqrt{2V\log(1/\delta)} + \frac{2b}{3}\log(1/\delta)$. In particular, for feedback Z_t with $\|Z_t\| \le B$ and $X_t = \langle Z_t - \mathbb{E}[Z_t \mid \mathcal{F}_{t-1}], w_t - u_t\rangle$ (u_t* predictable*, i.e. $\mathcal{F}_{t-1}$-measurable, so X_t is a martingale difference; $\|w_t - u_t\| \le D$), one may take $b = 2BD$ and $V = D^2\sum_t \mathbb{E}[\|Z_t\|^2 \mid \mathcal{F}_{t-1}]$.*

Proof. (X_t) is a martingale difference sequence ($\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0$). Freedman's inequality gives $\Pr(\sum_t X_t \ge \lambda) \le \exp(-\lambda^2/(2V + 2b\lambda/3))$; inverting the right-hand side to δ and using $\sqrt{x+y} \le \sqrt{x} + \sqrt{y}$ yields the bound. For the instantiation, $|X_t| \le \|Z_t - \mathbb{E}[Z_t \mid \mathcal{F}_{t-1}]\|\,\|w_t - u_t\| \le 2BD$ and $\mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}] \le D^2\mathbb{E}[\|Z_t\|^2 \mid \mathcal{F}_{t-1}]$ since $\mathbb{E}\|Z - \mathbb{E}Z\|^2 \le \mathbb{E}\|Z\|^2$. □

B.4. Varying-step telescope

Lemma B.4. *SN-OMD is OGD with step $\eta_t = \eta/s_t$ on the clipped raw gradient. For any comparator sequence $u_{1:T}$, writing $W_s = s_1 + \sum_t (s_t - s_{t-1})_+$ and $P_T^s = \sum_t s_t\|u_t - u_{t-1}\|$,*

$$\sum_{t=1}^{T} s_t\big(\|w_t - u_t\|^2 - \|w_{t+1} - u_t\|^2\big) \le D^2 W_s + 2D\,P_T^s.$$

Proof. Reindexing the sum and splitting each bracket into a comparator-move and a scale-change part,

$$s_t\|w_t - u_t\|^2 - s_{t-1}\|w_t - u_{t-1}\|^2 = s_t\big(\|w_t - u_t\|^2 - \|w_t - u_{t-1}\|^2\big) + (s_t - s_{t-1})\|w_t - u_{t-1}\|^2,$$

which is at most $2Ds_t\|u_t - u_{t-1}\| + (s_t - s_{t-1})_+ D^2$ by the reverse triangle inequality and $\|\cdot\| \le D$. Summing and dropping the nonpositive boundary term gives the bound. □

B.5. Dynamic regret (Theorem A.2)

Roadmap and where each assumption enters. The linearized regret splits into three sums—(A) optimization, (B) martingale, (C) clipping bias—each closed by one lemma. (A) uses the varying-step telescope (Lemma B.4) for the comparator/ scale-drift terms and the clipped second moment (Lemmas B.1 and B.2) for the curvature; (B) uses the Freedman step (Lemma B.3); (C) uses the clipping-bias bound (Lemma B.2). The two assumptions enter in exactly two isolated places: Assumption 2.1 (finite p-th moment) only through Lemma B.2, which converts $\mathbb{E}\|g_t\|^{p_t} \le \sigma_t^{p_t}$ into the conditional second-moment and bias bounds of (A)–(C); and Assumption A.1 (the lower bracket $s_t \ge c_1\sigma_t$) only in the curvature of (A)/(B) and the bias of (C)—*nowhere else*, which is precisely why over-estimating the scale is harmless. The dependency is *Lemma B.1*$\rightarrow$*Lemma B.2*$\rightarrow\{$(A),(B),(C)$\}$, with Lemma B.4 feeding (A) and Lemma B.3 feeding (B); the final display balances (A)'s curvature against (C)'s bias through the choice of η and M.

Write $d_t := \mathrm{clip}(g_t, Ms_t) = s_t\hat{g}_t$ for the clipped *raw* gradient, so that SN-OMD is exactly OGD with adaptive step $\eta_t = \eta/s_t$ on d_t (Lemma B.4). By convexity $R_T(u_{1:T}) \le \sum_t\langle\bar{g}_t, w_t - u_t\rangle$, and we split the true subgradient as

Table 2. Scale trackers on the real gradient-scale process (Section 4.3). The two-timescale envelope holds the lower bracket at 100% with the smallest upward variation W_s. V_σ^+ is the true upward scale drift.

Tracker	Lower bracket	W_s	W_s/V_σ^+
Trailing median, fast (W=64)	0.93	38.6k	60×
Trailing median, slow (W=500)	0.99	5.0k	7.9×
Two-timescale envelope (ours)	**1.00**	**4.2k**	**6.6×**

$\bar{g}_t = m'_t + \beta_t$, where $m'_t = \mathbb{E}[d_t \mid \mathcal{F}_{t-1}]$ and $\beta_t = \bar{g}_t - m'_t = \mathbb{E}[(g_t - d_t) \mid \mathcal{F}_{t-1}]$ is the (raw) clipping bias. Then

$$\sum_t \langle \bar{g}_t, w_t - u_t \rangle = \underbrace{\sum_t \langle d_t, w_t - u_t \rangle}_{\text{(A) optimization}} + \underbrace{\sum_t \langle m'_t - d_t, w_t - u_t \rangle}_{\text{(B) martingale}} + \underbrace{\sum_t \langle \beta_t, w_t - u_t \rangle}_{\text{(C) bias}}.$$

(A) The OGD one-step inequality with step η_t, summed and combined with Lemma B.4, gives $\sum_t \langle d_t, w_t - u_t \rangle \le \frac{1}{2\eta}(D^2 W_s + 2DP_T^s) + \frac{\eta}{2}\sum_t \|d_t\|^2/s_t$. Since $\|d_t\|^2 = \|\mathrm{clip}(g_t, Ms_t)\|^2 \le (Ms_t)^{2-p}\|g_t\|^p$ (Lemma B.1) and $s_t \ge c_1\sigma_t$ (Assumption A.1), the curvature sum has conditional mean $\le M^{2-p}\kappa' \sum_t \sigma_t = M^{2-p}\kappa' S_1$, with $S_1 = \sum_t \sigma_t$. *(B)* The increments $\langle m'_t - d_t, w_t - u_t \rangle$ form a martingale difference with $\|d_t\| \le Ms_t$ and $\mathbb{E}[\|d_t\|^2 \mid \mathcal{F}_{t-1}] \le M^{2-p}\kappa'\sigma_t^2$; Lemma B.3 bounds this sum by $\tilde{O}(DM^{1-p/2}\sqrt{\sup_t \sigma_t \cdot S_1})$, the same order as (A). This step requires u_t predictable (Lemma B.3), so the Freedman event is comparator-specific; Theorem A.2 is accordingly a guarantee *per fixed* $u_{1:T}$, not one holding uniformly over all comparators (which would need a union or chaining bound over comparator sequences, not pursued here). *(C)* $\|\beta_t\| \le \mathbb{E}[\|g_t\|^p \mid \mathcal{F}_{t-1}]/(Ms_t)^{p-1} \le \kappa'\sigma_t/M^{p-1}$ (Lemma B.2), so (C) $\le D\kappa' S_1/M^{p-1}$. Choosing $\eta = \Theta\big(\sqrt{(D^2 W_s + DP_T^s)/(M^{2-p}S_1)}\big)$ and $M = \Theta(T^{1/p})$ balances the curvature term of (A) against the bias (C) and yields

$$R_T(u_{1:T}) = O\Big(\sqrt{(D^2 W_s + DP_T^s)\, M^{2-p} S_1} + DS_1/M^{p-1}\Big),$$

which with $M = \Theta(T^{1/p})$ equals $(D\sqrt{W_s} + \sqrt{DP_T^s})\sqrt{S_1}\,T^{(2-p)/(2p)}$ up to the bias term. *If* the scale variation were bounded, $W_s = O(1)$, and using $S_1 \le T\sup_t \sigma_t$, this collapses to $\tilde{O}((\sup_t \sigma_t)(D + \sqrt{DP_T})T^{1/p}\log(1/\delta))$—but W_s is a data quantity, and we *measure* $W_s \sim T$ (Section 4.3). So the $\sqrt{W_s}$ factor is a genuine $\sqrt{T}$ rate and this $T^{1/p}$ simplification holds only *per regime*; globally the bound is $T^{1/p+1/2}$ (at $p = 2$, the vacuous T). The stationary case ($\sigma_t \equiv \sigma$, $p = 2$, no drift so $W_s = \sigma$) recovers $\sigma(D + \sqrt{DP_T})\sqrt{T}$, the classical dynamic-OGD rate. □

B.6. Reductions

Unknown p: $M = \Theta(T^{1/p})$ needs p; a doubling schedule over M, using that the bound is unimodal in $\log M$ with a flat optimum, removes this at a $\mathrm{polylog}(T)$ cost (Zhang et al., 2026). *Unknown $\|u\|$ / unbounded domain:* Theorem 3.1 gives bounded feedback $\|\hat{g}_t\| \le M$, exactly the input required by the parameter-free potential of Zhang & Cutkosky (2022); substituting our $\hat{g}_t$ yields the unbounded-domain analogue with D replaced by $\|u\|$. *Unknown regime length:* Lemma B.4 plus the interval version of Lemma B.3 give SN-OMD an interval-regret guarantee on every $[a, b]$; a strongly-adaptive wrapper (Daniely et al., 2015; Cutkosky, 2020) then recovers the dynamic bound without knowing the regime structure, at a $\mathrm{polylog}$ overhead. A per-regime transition cost $O(D\,N\,\mathrm{polylog}\,T^{1/p})$ is incurred while the scale tracker re-concentrates after each of the N regime changes; it is dominated when regimes are macroscopic.

B.7. Discharging Assumption A.1: a bracket guarantee

Assumption A.1 posits a predictable scale that brackets σ_t. We show it is not an oracle: under a mild model of the drift, the two-timescale envelope of Section 4.3 *provably* meets the essential lower bracket off a small exceptional set, so the burden moves from "an accurate scale estimate exists" to "the drift is piecewise slowly varying," and the estimate is then *derived* from standard robust concentration.

Drift model. Partition $[T]$ into N regimes by change points $t_0 < \cdots < t_N$. *Slow within-regime drift:* inside a regime the scale varies slowly, $\sigma_t/\sigma_{t-1} \in [1-\gamma, 1+\gamma]$ with $\gamma W \le c_0$ for the window W. *Arbitrary jumps:* at the N change points σ_t may jump by any factor. This is the piecewise-slowly-varying picture the measured gradient scale fits (∼30% within-day drift, occasional 3× overnight jumps; Section 4).

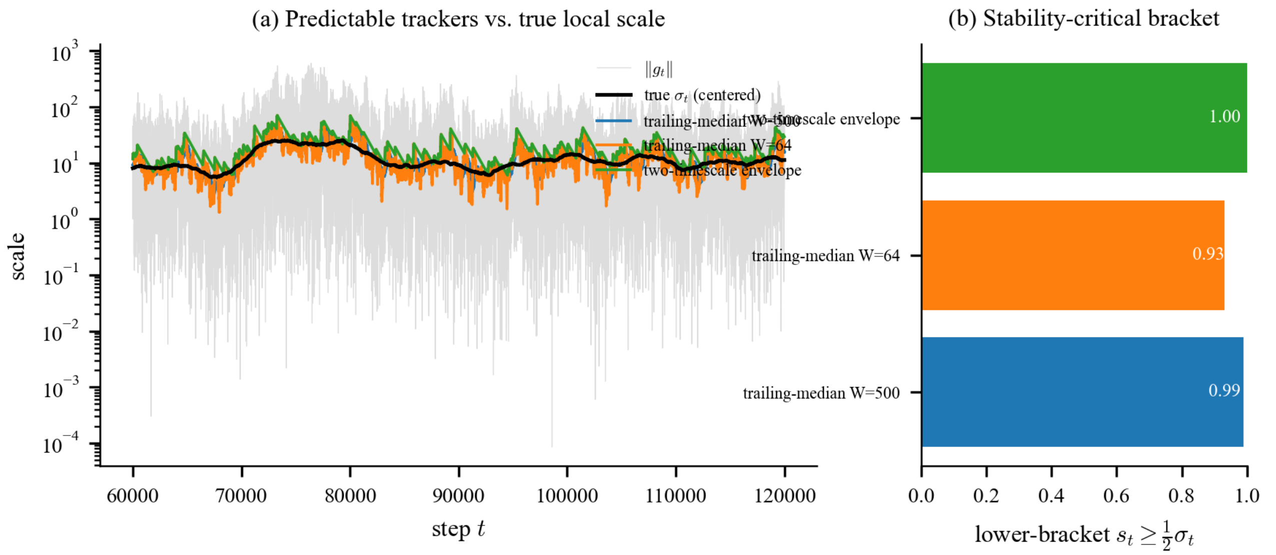

Figure 6. **Discharging Assumption A.1** (Section 4.3). A two-timescale "peak-hold" tracker on the real gradient-scale process (a) holds the lower bracket $s_t \geq \frac{1}{2}\sigma_t$ at every round while (b) attaining the smallest upward variation W_s/V_σ^+; a fast trailing median churns (60×), a slow one lags and violates the bracket.

Table 3. The scale drift is a genuine rate, but its *constant* is controllable. Upward variation W_s vs. the tracker's update block length B (predictable robust median per block) on the 200,000-round gradient-scale process. Restricting the $W_s \sim T^\beta$ fit to horizons with $\gtrsim 50$ blocks gives $\beta \approx 1.0$–1.25 where reliably measurable (B=100, 10^3), with no monotone trend; the naive full-range fit's decline (to 0.17 at large B) is a finite-horizon artifact—B=10^4 has only $\sim$20 blocks and no $\geq$50-block horizon. What B controls is the constant, shrinking W_s by $\sim$5 orders of magnitude while roughly holding the lower bracket. Escaping the resulting $\sqrt{T}$ drift factor needs a switching bound in the regime count, left open.

block B	1	100	10^3	$3{\cdot}10^3$	10^4
W_s (full record)	1.8M	3216	319	108	29
β ($\geq$50 blk)	—	1.03	1.25	0.73	n/a
lower bracket	0.72	0.97	0.97	0.92	0.93

Tracker. The envelope $s_t = \max\{c\,\widehat{m}_t,\,(1-\rho)s_{t-1}\}$, where $\widehat{m}_t = \operatorname{median}(\|g_{t-W}\|,\ldots,\|g_{t-1}\|)$ is the predictable short-window median and $\rho \asymp 1/L$ with $L = \min_r L_r$.

Proposition B.5 (Bracket guarantee)**.** *Under Assumption 2.1 and the drift model above, for a large enough constant c, with probability $\geq 1-\delta$:* (i, lower bracket) $s_t \geq c_1\sigma_t$ *for every t outside at most $O(W)$ rounds after each jump—an exceptional set of size $O(NW)$; and* (ii, bounded variation) $W_s = O(\sup_t \sigma_t + \rho\sum_t \sigma_t) = O(V_\sigma^+)$. *Consequently Theorem A.2 holds with one added term $O(D\,N\,W\,\sup_t\sigma_t)$, dominated whenever $NW \ll T^{1/p}$ (macroscopic regimes).*

Proof sketch. Window concentration. On a window lying inside one regime, the population median of $\|g_t\|$ is a constant multiple of σ_t (regular variation), and the empirical median concentrates around it by order-statistic bounds (needing essentially no moment); the slow drift shifts the constants by $O(\gamma W) = O(1)$. Hence $\widehat{m}_t \in [a\sigma_t, b\sigma_t]$ w.h.p., and $c \geq c_1/a$ gives $c\,\widehat{m}_t \geq c_1\sigma_t$, so the $\max$ meets the lower bracket whenever the window lies in one regime. *Jumps.* After an upward jump the window still holds pre-jump norms for $\leq W$ rounds; during that lapse the median term may under-estimate, giving the $O(NW)$ exceptional set. A downward jump only makes s_t conservatively high (the release lets it fall at rate ρ), never violating the lower bracket. *Variation.* Upward moves of s_t come only from $c\,\widehat{m}_t$ rising; between jumps these telescope to $\sup_t s_t$, and the geometric release contributes the churn $\rho\sum_t s_t$, so $W_s \leq \sup_t s_t + \rho\sum_t s_t$, which is $O(V_\sigma^+)$ for $\rho \asymp 1/L$ (Remark A.4). *Exceptional rounds.* On the $O(NW)$ lapse rounds the update is still bounded ($\|\hat{g}_t\| \leq M$, Theorem 3.1) but the clipping bias (C) is uncontrolled; charging it trivially at $O(D\sup_t\sigma_t)$ per round gives the added $O(D\,N\,W\,\sup_t\sigma_t)$. □

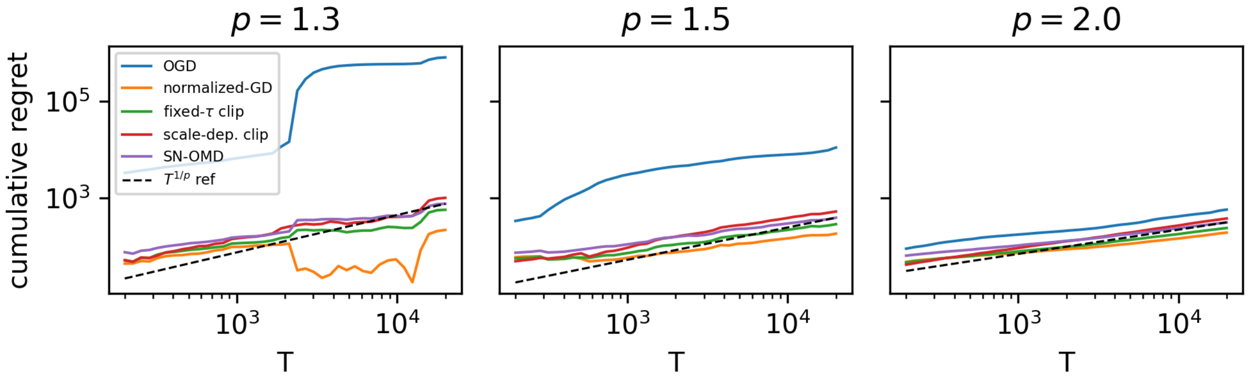

Figure 7. **Known-tail synthetic** (Section 4.2). Cumulative regret vs. T at $p \in \{1.3, 1.5, 2.0\}$: OGD's exponent climbs past 1 (divergent) as the tail heavies, while SN-OMD stays sublinear and below the $T^{1/p}$ reference (dashed).

The residual—the post-jump lapse—is an additive term already of the form isolated in the Reductions above, and making ρ adaptive to an unknown L is exactly the strongly-adaptive wrapper of that subsection.

B.8. The deployed EMA vs. the analyzed envelope tracker

The regret analysis discharges Assumption A.1 with the two-timescale "peak-hold" envelope of Section 4.3, whereas the empirically strongest configuration uses a winsorized exponential moving average (EMA). Because a reviewer may reasonably ask whether the analyzed object and the deployed object are the same, we state precisely what does and does not depend on the choice of tracker.

Stability is tracker-independent. Theorem 3.1 uses only $\|\hat{g}_t\| \leq M$, which the cap enforces for *any* predictable $s_t > 0$. Non-divergence therefore holds verbatim for the EMA, for the envelope, and for any other tracker; the gap discussed here is never about stability, only about the leading regret constant.

Only the regret constant depends on the tracker. For any predictable s_t meeting the essential lower bracket $s_t \geq c_1 \sigma_t$, the rate is $T^{1/p}$ (Theorem A.2); trackers differ only through the leading factor $\sqrt{W_s}$. The envelope is the tracker for which we can *prove* both the lower bracket and $W_s = O(V_\sigma^+)$ (Proposition B.5)—reacting up on a short robust median and releasing down only geometrically keeps every upward move a genuine rise (Table 2: 100% lower bracket, $W_s = 6.6\, V_\sigma^+$). A fast, *unwinsorized* EMA can in principle churn on heavy-tailed $\|g_t\|$ spikes (Table 2: $W_s = 60\, V_\sigma^+$); winsorizing the EMA at $k \cdot \mathrm{median}$ caps each upward step and is what keeps the deployed default's W_s finite in practice, though we do not give it a closed-form bound.

Which tracker to deploy: a three-way gap, widest at the best point. There are now three trackers in play and the accuracy/guarantee trade-off has three corners. *(1) The envelope* $s_t = \max\{c\,\hat{m}_t, (1-\rho)s_{t-1}\}$ is the only one with a *proven* bound: Proposition B.5 gives it both the lower bracket and $W_s = O(V_\sigma^+)$. *(2) The winsorized EMA* (deployed default) has no closed-form W_s bound—Proposition B.5 does not cover it, as its functional form is not the envelope's—only the empirical $W_s = 60\, V_\sigma^+$ of Table 2. *(3) The block median* ($B{=}10^4 \approx$ one day) is the most accurate *per-row* on our stream (per-timestep the constant-τ clip edges it, 0.382 vs. 0.351; Table 1) yet the least analyzed: Proposition B.5 does not cover it either, and we prove nothing about its W_s. So the best-performing tracker is the one with no guarantee; the gap between what we can *prove* (envelope) and what we would *deploy* (EMA, or block) is if anything wider than before, and stating that cleanly is more honest than implying it narrowed.

On accuracy we are deliberately careful, because the numbers are easy to over-read (paired 10-fold bands, each method at its tuned rate; Section 4.3). The block tracker *does* beat the normalized-GD baseline per-row—0.38 vs. 0.197, $t \approx 10.6$, all 10 folds—but that edge is *protocol-specific*: under the batched per-timestep protocol it vanishes (0.351 vs. 0.316, $t \approx -0.1$). The mechanism is substitution, not coincidence: per-timestep batching averages the ~ 9 correlated symbols in each $(\mathrm{date}, \mathrm{time})$ group, and that averaging is itself the variance reduction the smooth block scale supplies per-row, so the two do not compound—when the protocol smooths the update, a smoother tracker adds nothing. The same averaging is what lifts normalized-GD ($0.197 \to 0.316$, it had no such smoothing per-row) while, uncapped, it *enlarges* the untruncated step and halves scale-adaptive OGD ($0.162 \to 0.080$, Section 4); the block tracker's edge sitting exactly where the protocol lacks the smoothing is the same story from the tracker side. Against the EMA *default* its edge is *not* significant on either protocol ($+0.16$ per-row but $t \approx 1.3$, $8/10$ folds). So we do *not* claim the block tracker is uniformly more accurate, and we keep the winsorized EMA as the reported default: it is the standard reactive choice, and the block tracker does not beat it beyond fold noise. Past the tuned range every tracker *degrades in loss* without diverging in the sense of Theorem 3.1—at lr=10 the peak rolling loss reaches ≈ 30 (block) or ≈ 230 (EMA) while the $O(\sqrt{t})$ iterate bound still holds—so the earlier "pick either without risking divergence" was loose: the iterate never blows up, but usability does past tuning, the block tracker most gracefully.

This may look in tension with Section 4.3/Figure 4, where long windows are penalized for lagging regime onsets. The tension is only apparent, and it is exactly the distinction between the two uses of the scale. When the scale is a *divisor* ($\hat{g}_t = g_t/s_t$), a lagging s_t merely mis-normalizes the step magnitude—the divisor cancels first-order and a slightly stale scale costs little, so a slow block median is tolerable and even helpful (it is smoother and higher-bracket). We measured this rather than asserting it: sweeping the tracker's block timescale over $B \in \{50, \ldots, 20{,}000\}$ (a $400\times$ window range) moves the divisor's best-tuned weighted R^2 only within $[0.363, 0.386]$ (spread 0.022), if anything *rising* with the window—so a longer, laggier scale does not hurt the normalizer. When the scale is instead a *threshold* (clipping $\|g_t\|$ at s_t), the same sweep says something *sharper* than "long windows hurt thresholding." The threshold use never leaves $R^2 \approx 0.003$ at *any* window: clipping $\|g_t\|$ at a tracked scale is scale-*dependent* OGD, stable only at a tiny step (lr $\approx 3 \times 10^{-4}$), so it is unusable no matter how the scale is tracked. Window choice is therefore moot on the threshold side—there is no usable operating point to tune toward—while the divisor is usable and window-insensitive across the whole $400\times$ range. This is the paper's central fragility finding (Section 4) arriving from the tracker-design direction: normalization decouples the step from the scale and thresholding does not, so the very lag that would wreck a threshold (it clips hardest exactly when the scale jumps up, discarding the informative post-onset gradients—the Section 4.3 clip-rate/dynamic-range tension) merely re-normalizes a divisor. The one-day block tracker is a normalizer, so no contradiction arises.

Closing the gap. Either run the envelope in deployment, accepting the small accuracy cost, or prove a W_s bound for the winsorized EMA under a heavy-tailed scale process: the winsorization already supplies the per-step cap the envelope gets from its median, so what remains is to control the geometric-decay churn term $\rho \sum_t s_t$, exactly as in Remark A.4, with ρ the EMA rate. The one residual modeling assumption—ρ matched to an unknown regime length—is removed by the strongly-adaptive wrapper of the preceding subsection.